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102 ①

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Machine Learning

03- Scaling: Min-Max and Standard

A) ①

[18, 42, 66, 90]

$$\min(x) = 18$$

$$\max(x) = 90$$

$$90 - 18 = 72$$

$$[0, 0.33, 0.66, 1]$$

$$\frac{42 - 18}{72} = 0.33$$

$$\frac{66 - 18}{72} = 0.66$$

B) [8, 10, 10, 14, 14, 16] $n = 6$

$$\text{Mean} = \frac{8 + 10 + 10 + 14 + 14 + 16}{6} = 12$$

$$\text{Std} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

$$\text{Std} = \sqrt{\frac{(8-12)^2 + (10-12)^2 + (10-12)^2 + (14-12)^2 + (14-12)^2 + (16-12)^2}{6}}$$

$$② = \sqrt{\frac{16 + 4 + 4 + 4 + 4 + 16}{6}} = \sqrt{\frac{48}{6}} = \sqrt{8} = 2.828$$

$$\bar{x} = 12 \text{ every } \text{div} = [-4, -2, -2, 2, 2, 4] \text{ Std} = \sim 2.83$$

① C)

②

③ Min-Max:- The advantage is that it gives a scale between $[0, 1]$ which is a fixed range.
- The disadvantage is that one extreme value can change the whole scaler

Standardisation:- The advantage is that it is less affected by extreme values
- The disadvantage is that there is no fixed range so values can be large or negative.